BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 134 · Lesson 134 · Database Foundations: Flat Files, Tables and Keys

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a cryptographic digest help verify?

Q2. Which is removed first from a FIFO queue?

Q3. What is a composite key?

Q4. What are the outputs of a half adder when A=1 and B=1?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

record — A row representing one entity instance.

primary key — Unique, non-null identifier for a table row.

foreign key — Column referencing a key in another table.

flat file — Data kept in one simple table or file without relational links.

New learning

Single flat file — Duplicating pupil details on each enrolment creates repetition.

Relational tables — Pupil and Enrolment records separate into related entities.

Keys connect — Primary key identifies a row; foreign key preserves relation to parent.

Worked example

A pupil books several clubs. Why store pupils and club bookings in separate tables?

1. Pupil ID 7: Sam — Find the given data and name the required result.

2. Bookings refer to PupilID 7 — Explain why this intermediate stage is necessary.

3. Change name once — Pupil(PupilID PK, Name) and Booking(BookingID PK, PupilID FK, ClubID FK) separate data. A pupil name is held once and each booking refers to the pupil by ID, reducing update anomalies.

We Do · Practise with guidance

A school has Device(DeviceID, Model) and Loan(LoanID, DeviceID, Borrower). Identify suitable keys.

Device PK DeviceID

Find the given data and name the required result.

Loan PK LoanID

Explain why this intermediate stage is necessary.

Loan FK DeviceID

DeviceID uniquely identifies each device (PK). LoanID identifies each loan (PK). Loan.DeviceID is an FK to Device.DeviceID. Borrower may later reference a Borrower table.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define primary key and foreign key using pupil/club tables. [6]

Question 2

Explain two problems with keeping pupil contact details on every booking row. [6]

Question 3

Design a relational structure for Customers placing multiple Orders, showing suitable PKs and FK. [8]

Exit Ticket · Answer before checking

A foreign key value must always be unique in its table.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Integrity, when compared against a trusted expected digest.

Q2. The longest-waiting item at the front.

Q3. A key made from two or more attributes together.

Q4. Sum 0, carry 1.

You Do pages · Independent questions

Q1. Unique PK [2], FK references primary/candidate key in another table [2], valid specific column examples [2].

Q2. Duplication wastes storage [2], inconsistent updates [2], insert/delete anomalies [2].

Q3. Customer(CustomerID PK,...)[2], Order(OrderID PK,...)[2], Order.CustomerID FK[2], accurate one-to-many/benefits [2].

Exit Ticket · Model answer

False. Many child records may share one valid parent key.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: